DATA ANALYTICS / BANGLADESH

Food price intelligence & risk analysis

From raw market observations to a decision-focused commodity monitoring system.

34,490	43 / 66	May 2026
Clean records	Eligible risk series	Comparison month

The business problem

A food retailer or distributor needs early signals about unusual prices and where to recheck suppliers. Raw market data mixes different products, market locations, measurement units and time periods, so naive averages can mislead decision-makers.

What I built

A reproducible Python → SQL pipeline with quality checks, matched-market monthly comparisons, fixed-panel risk scoring, an interactive dashboard and an independently validated results layer.

Critical interpretation note

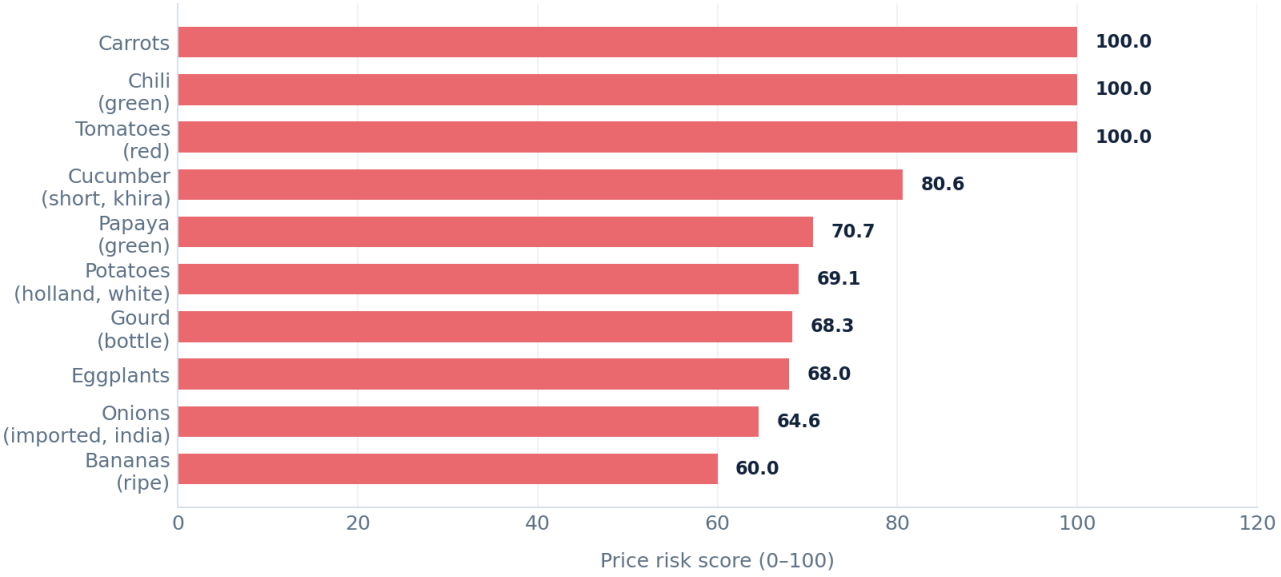
The 12-month risk watchlist represents four markets in the Chittagong area. Its ranks are local warnings, not national commodity rankings. September 2026 is the latest source month; May 2026 is the last month meeting the pipeline's comparison coverage rule.

SOURCE: WFP Bangladesh food prices via the Humanitarian Data Exchange (HDX). This is nominal BDT price analysis, not CPI or a causal impact assessment.

01 / SIGNAL DETECTION

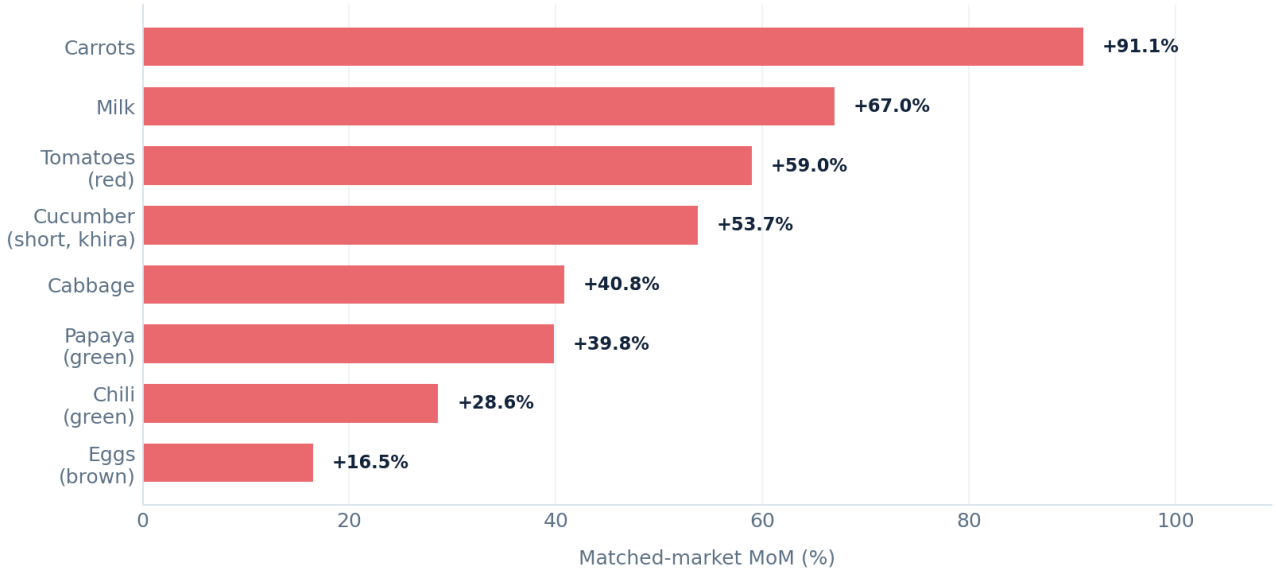
Where prices need attention

Each score is computed for one commodity, unit and price type, using twelve consecutive months with identical panel markets.



Top-ranked carrots, green chili and red tomatoes reach the 100-point cap; equal capped values do not imply equal risk. Each of the top-ranking series requires careful review of panel sizes, source flags, and local applicability.

Latest matched-market monthly increases

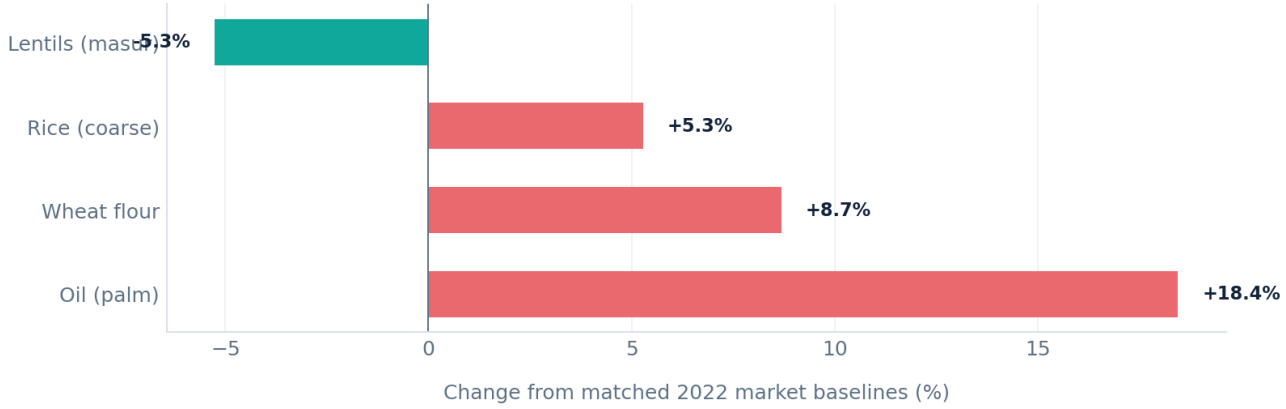


May 2026 observed sharp matched-market increases for carrots, milk and red tomatoes. Results reflect eligible same-market comparisons, not a national inflation index.

02 / STAPLE CONTEXT

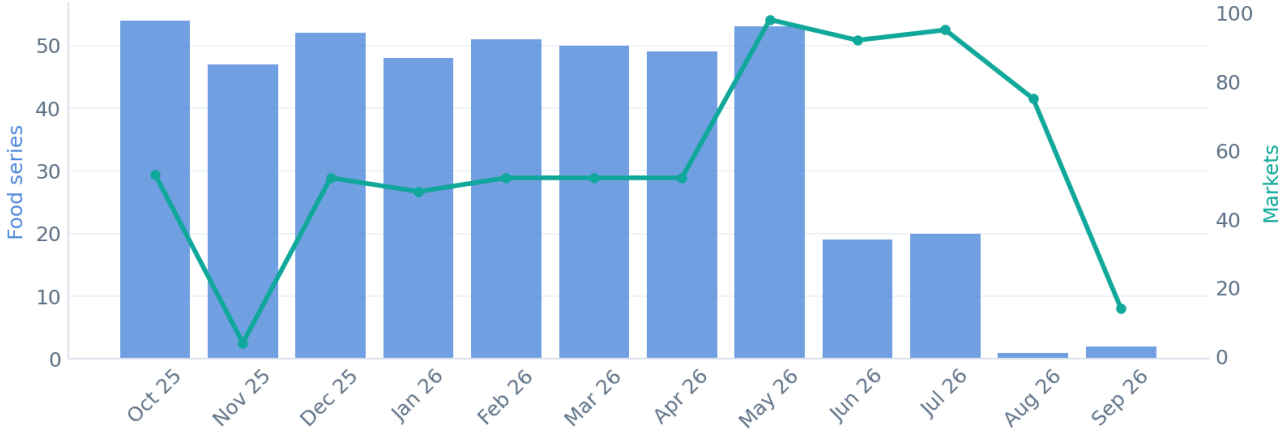
Longer-term prices and data coverage

Matched markets reduce composition bias; they do not remove measurement differences between source aggregates and direct observations.



Compared with matched market-level 2022 baselines, palm oil was approximately +18.4%, wheat flour +8.7%, coarse rice +5.3% and masur lentils -5.3% at the May 2026 snapshot. Baselines use available months and include source aggregate flags.

Why the reference month is not the latest source month



The pipeline selects the latest month reaching at least half the trailing-year maximum for both food-series counts and market counts. Later months are retained in source coverage and trends but not used for the ranked snapshot. This is a documented heuristic, not proof that any month is complete.

03 / REPRODUCIBLE PIPELINE

From data to a decision

An analytical workflow designed to make the output auditable, explainable, and repeatable.

01 Ingest and profile

Download the original WFP CSV manually. Profile columns, date coverage, unit labels, price types, missing values and measurement flags. Preserve the original source unchanged.

02 Clean and validate

Standardize fields, check dates, prices and currencies, remove exact duplicates, quarantine invalid records, and keep suspicious outliers with flags rather than silently removing them.

03 Build the SQL model

Load fact prices and commodity, market and calendar dimensions into SQLite or PostgreSQL. Execute 12 SQL analyses and export the explicit comparison cohorts.

04 Calculate careful comparisons

Use endpoint-matched markets for MoM/YoY. Use identical market membership for all 12 months of the risk panel. Never mix unlike measurement units or wholesale and retail.

05 Verify and present

Run automated tests plus independent calculations, then generate findings, static figures, an interactive offline dashboard and this case-study report.

Data quality is part of the finding

34,491 raw records become 34,490 cleaned observations; 1 record is quarantined. Of 66 candidate food-risk series, 43 qualify for the strict panel. Excluded series are **unscored**, not low risk.

04 / ACTIONS & LIMITS

Recommended business decisions

The dashboard supports prioritized investigation; it does not substitute for procurement verification or a forecasting model.

Priority 1 — check local suppliers

For high-scoring products, compare current quotations across the same local markets. Reconfirm product grade, season, transportation cost and unit before changing purchasing terms.

Priority 2 — monitor before committing

Follow matched-market monthly increases and price dispersion. For produce with sharp spikes, schedule earlier checks; do not automatically bulk-buy from a capped risk score.

Priority 3 — broaden market evidence

Before making national distribution decisions, add broader geographic coverage and verify measurement consistency across WFP releases.

Priority 4 — operationalize refresh

Run the Windows batch file when a new WFP CSV is available. Revalidate records, update the analysis month selection, rerun comparisons and regenerate dashboard files.

What this report does not claim

No official national inflation series, no validated price forecast, no causal proof, no verified procurement savings, no assumption that ineligible markets are safe, and no claim that the dashboard's four-market risk score describes the whole country.

Project stack

Python / pandas for profiling and cleaning • **SQL** for monthly comparisons and ranking • **SQLite / PostgreSQL** for reproducible storage • **Plotly** for interactive reporting • **Power BI** build assets for Windows Desktop.

Source and reproducibility

Source: World Food Programme (WFP), Bangladesh Food Prices, distributed through HDX. Open source dataset listing. License for the precise downloaded release must be verified before redistributing the source dataset. Code, queries and validation controls are documented in the project repository.

This case study is an independent portfolio exercise and is not affiliated with or endorsed by WFP or HDX.